# Tomatoes

### 1. Late Blight (*Phytophthora infestans*)

- **Description:** Late blight is a fast-spreading fungal-like disease affecting leaves, stems, and fruits. Infected areas become dark, water-soaked, and may rot quickly. Humid and wet conditions accelerate its spread.
- **Guidance:** Remove and destroy infected plants immediately. Space plants 50–70 cm apart for better airflow. Avoid planting tomatoes after potatoes in the same soil.
- **Irrigation:** Keep soil evenly moist without wetting leaves. Water 2–3 cm per week, preferably early morning via drip irrigation.
- **Fertilization:** Apply nitrogen 120 kg/ha, potassium 50–60 kg/ha, and phosphorus 20–30 kg/ha in split doses. Avoid excess nitrogen during outbreaks.
- **Treatment:** Apply copper-based or systemic fungicides every **7–10 days** when wet conditions persist.

### 2. Early Blight (*Alternaria solani*)

- **Description:** Causes circular brown spots with concentric rings on older leaves. Leads to leaf drop and reduced photosynthesis, weakening the plant.
- **Guidance:** Remove infected leaves and mulch soil to prevent splash infections. Space plants 50–60 cm apart.
- **Irrigation:** 2 cm every 2–3 days; water early morning and avoid wetting foliage.
- **Fertilization:** 100 kg/ha nitrogen + 50 kg/ha potassium in 2–3 split applications.
- **Treatment:** Apply chlorothalonil or mancozeb every **10–14 days** during wet periods.

**3. Tomato Mosaic Virus (ToMV)**

- **Description:** Virus causes mottled yellow leaves, leaf distortion, and reduced fruit quality. Spread through contaminated tools, hands, and seeds.
- **Guidance:** Remove infected plants immediately. Use virus-free seeds and clean tools with 70% alcohol. Avoid handling plants when wet.
- **Irrigation & Fertilization:** Standard care; avoid excess nitrogen.
- **Treatment:** No chemical cure; prevention is essential.

**4. Fusarium Wilt (*Fusarium oxysporum*)**

- **Description:** Soil-borne fungus causing yellowing, wilting, and stunted growth, usually starting on one side of the plant. Severe infections can kill plants.
- **Guidance:** Plant resistant tomato varieties. Solarize soil for 4–6 weeks before planting. Ensure good drainage.
- **Irrigation:** Drip irrigation, 2–3 cm/week; avoid waterlogging.
- **Fertilization:** Moderate potassium (60 kg/ha). Avoid excess nitrogen.
- **Treatment:** Prevention through crop rotation and resistant varieties.

# Potatoes

**1. Late Blight (*Phytophthora infestans*)**

- **Description:** Dark lesions on leaves and tubers. Rapidly spreads in cool, wet conditions, causing severe yield loss.
- **Guidance:** Remove infected foliage and tubers. Rotate crops every 3–4 years. Plant resistant varieties.
- **Irrigation:** Avoid wetting foliage; drip irrigation preferred.
- **Fertilization:** N 120 kg/ha, K 150 kg/ha, P 60 kg/ha in split doses.
- **Treatment:** Copper fungicide or mancozeb every **7–10 days** during wet weather.

**2. Early Blight (*Alternaria solani*)**

- **Description:** Circular dark spots on leaves, reducing photosynthesis and yield. Affects older leaves first.

- **Guidance:** Remove infected leaves; mulch to prevent soil splash. Space plants for airflow.
- **Irrigation:** 2–3 cm/week, avoid wetting foliage.
- **Fertilization:** Balanced NPK 120:60:60 kg/ha.
- **Treatment:** Chlorothalonil every **10–14 days**.

**3. Potato Virus Y (PVY)**

- **Description:** Mottled leaves, leaf drop, poor tuber development. Spread mainly via aphids and infected seed tubers.
- **Guidance:** Use certified virus-free seed tubers. Control aphids with neem oil or insecticides weekly. Remove infected plants.
- **Irrigation & Fertilization:** Standard potato requirements.
- **Treatment:** No chemical cure; focus on vector and seed control.

**4. Common Scab (*Streptomyces scabies*)**

- **Description:** Rough, corky lesions on tubers. Favored by dry soils, does not usually affect plant growth.
- **Guidance:** Rotate crops and avoid planting in dry soils during early tuber formation.

- **Irrigation:** Maintain slightly acidic soil (pH 5.0–5.2); water 2–3 cm/week.
- **Fertilization:** Avoid excess nitrogen; potassium 60–80 kg/ha helps reduce lesions.
- **Treatment:** Prevention through irrigation management and crop rotation.

# Apples

**1. Apple Scab (*Venturia inaequalis*)**

- **Description:** Dark velvety spots on leaves and fruits. Causes leaf drop and reduces fruit quality. Favors wet, cool spring weather.
- **Guidance:** Remove fallen leaves, prune for airflow. Plant resistant varieties.
- **Irrigation:** Keep soil moist (2–3 cm/week), avoid wetting leaves.
- **Fertilization:** N 80–100 kg/ha, K 40–60 kg/ha in split applications.
- **Treatment:** Captan or myclobutanil fungicide every **10–14 days** in wet periods.

**2. Fire Blight (*Erwinia amylovora*)**

- **Description:** Bacterial disease blackening blossoms, shoots, and branches. Rapidly kills young branches.

- **Guidance:** Prune infected branches in dry weather, disinfect tools. Plant resistant varieties. Avoid excess nitrogen.
- **Irrigation:** Moderate; avoid wetting flowers in spring.
- **Fertilization:** K 50–60 kg/ha.
- **Treatment:** Apply streptomycin at **early bloom**, repeat if rain occurs.

**3. Powdery Mildew (*Podosphaera leucotricha*)**

- **Description:** White powdery growth on leaves, shoots, and buds; reduces growth and fruit quality.
- **Guidance:** Prune crowded shoots; remove infected shoots. Plant resistant varieties.
- **Irrigation:** Avoid wet foliage; 2–3 cm/week at soil.
- **Fertilization:** Moderate nitrogen; K 40–60 kg/ha.
- **Treatment:** Sulfur or systemic fungicide every **10–14 days**.

**4. Cedar-Apple Rust (*Gymnosporangium juniperi-virginianae*)**

- **Description:** Orange or yellow spots on leaves and fruit; requires juniper trees as alternate hosts.
- **Guidance:** Remove nearby junipers if possible. Plant resistant varieties.
- **Irrigation:** Standard, avoid wet foliage.
- **Fertilization:** N 70–90 kg/ha, K 40–50 kg/ha.
- **Treatment:** Apply mancozeb at **bud break**, repeat every 10 days.